OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester I Examinations 2024-25

Exam Code(s)	4BMS2, 4BS2, 4BPT2, 4PHO1, 1HA1, 1AL1
Exam(s)	4th Science
Module(s)	Algebraic Foundations of Quantum Computing
Module Code(s)	MA4102
External Examiner(s)	Prof. Colva Roney-Dougal
Internal Examiner(s)	Dr. Niall Madden
	*Dr. Michael Mc Gettrick
Instructions	Answer **FOUR** questions.
Duration	2 hours
No. of Pages	4 pages (including this cover page)
Discipline	Mathematics

Requirements:

Release to Library	Yes	☑	No	☐
MCQ	Yes	☐	No	☑
Statistical/Log Tables	Yes	☑	No	☐

1. (i) Consider the **qutrit** state given by

$$|\psi\rangle = \frac{1}{3}\left(-2\,|0\rangle - i\,|1\rangle + 2i\,|2\rangle\right).$$

(a) What is the probability of measuring the particle in the $|2\rangle$ state?

(b) Calculate the density matrix ρ_ψ.

(c) If we operate on $|\psi\rangle$ using the unitary

$$U = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix},$$

calculate the subsequent probability of measuring $|2\rangle$.

[14 marks]

(ii) Consider the **qubit** state

$$|\psi\rangle = \frac{1}{2}\left(\sqrt{3}\,|0\rangle + \frac{(2+i)}{\sqrt{5}}\,|1\rangle\right).$$

(a) If $|\psi\rangle$ is represented on the Bloch Sphere, calculate the corresponding angles θ and ϕ.

(b) If we measure using the orthonormal basis $|+\rangle$ and $|-\rangle$, calculate the probability of obtaining $|+\rangle$.

[11 marks]

2. (i) Suppose we write a bipartite (2-qubit) state as a column vector with 4 components, using the standard basis $|00\rangle = (1,0,0,0)^T$, etc. Calculate (a) the purity and (b) the entanglement (using the Von Neumann entropy of the reduced density matrix) for the second qubit for each of the following states.

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \\ 0 \\ 0 \end{pmatrix} \qquad |\psi_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ i \\ -i \\ 0 \end{pmatrix} \qquad |\psi_3\rangle = \frac{1}{\sqrt{10}}\begin{pmatrix} \sqrt{2} \\ \sqrt{3} \\ \sqrt{4} \\ i \end{pmatrix}$$

[18 marks]

(ii) Prove that Unitary transformations preserve state purity, i.e. if ψ is a quantum state and U is a Unitary transformation, then ψ and $U\psi$ have the same purity.

[7 marks]

3. (i) Given the orthonormal basis

$$|g\rangle = \frac{1}{\sqrt{7}}|0\rangle + i\frac{\sqrt{6}}{\sqrt{7}}|1\rangle, \qquad |h\rangle = \frac{-\sqrt{6}}{\sqrt{7}}|0\rangle + i\frac{1}{\sqrt{7}}|1\rangle$$

calculate the associated Kraus operators M_g and M_h for the amplitude damping channel U given by

$$U(|00\rangle) = |00\rangle \qquad U(|11\rangle) = |11\rangle$$
$$U(|01\rangle) = \sqrt{\frac{2}{3}}|01\rangle + \sqrt{\frac{-1}{3}}|10\rangle$$
$$U(|10\rangle) = \sqrt{\frac{-1}{3}}|01\rangle + \sqrt{\frac{2}{3}}|10\rangle$$

(where the second qubit represents the environment). **[13 marks]**

(ii) For the positive operator-valued measure (POVM) given by

$$E_1 = \frac{1}{2}|1\rangle\langle 1|, \qquad E_2 = \frac{1}{3}|+\rangle\langle +|, \qquad E_3 = 1 - E_1 - E_2,$$

calculate the probability of each of the three outcomes when measuring the state

$$|\psi\rangle = \frac{2 - i\sqrt{5}}{5}|0\rangle + \frac{4}{5}|1\rangle.$$

[12 marks]

4. (i) Prove the *no cloning theorem*, i.e. that it is impossible to copy a quantum state. **[7 marks]**

(ii) Calculate the Quantum Fourier Transform of the following qudit states:

$$\text{(a)}\ \frac{1}{\sqrt{3}}\begin{pmatrix}1\\ i\\ -i\end{pmatrix} \qquad \text{(b)}\ \frac{1}{\sqrt{3}}\begin{pmatrix}i\\ 1\\ -1\\ 0\end{pmatrix}$$

[10 marks]

(iii) Draw the circuit which implements quantum teleportation, i.e. where Alice uses a maximally entangled qubit pair, shared with Bob, to transmit an unknown qubit state ψ to Bob. **[8 marks]**

5. Suppose we are carrying out a Grover search of a list of length 32, for an item X which is guaranteed to occur exactly once.

   (i) Calculate the theoretical number of Grover iterates we should apply (this need not be a Natural Number). **[6 marks]**

   (ii) Calculate how many (integer) iterations we should do to maximize our success probability, and calculate the success probability at that point. **[11 marks]**

   (iii) How many iterations will give us the "second best" chance of finding X, and what is the associated probability? **[8 marks]**